PROBLEM FS3

Hollow Cube with Trapped Air

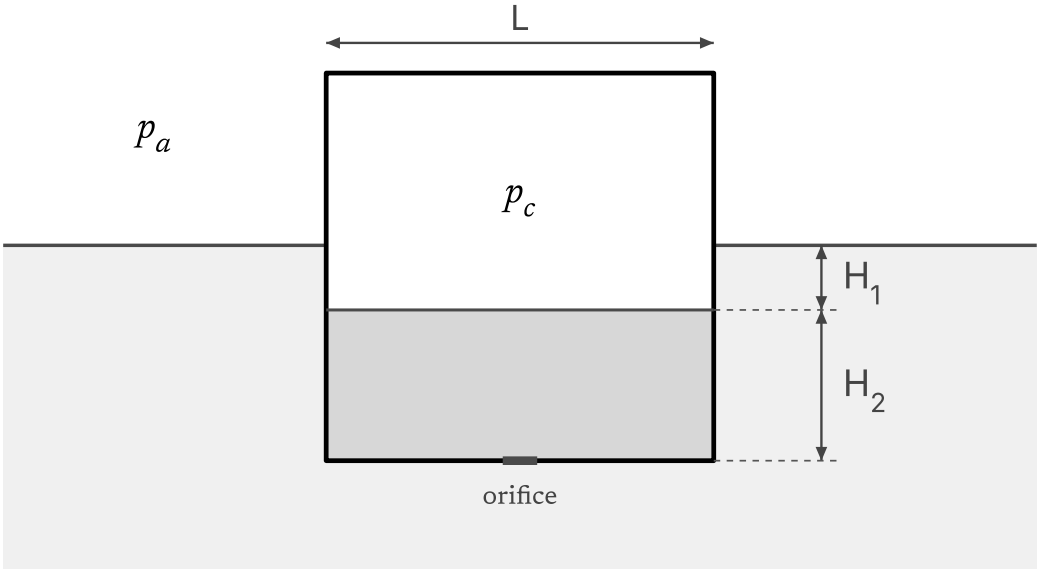
Original statement · fluid_statics_3.pdf

Floating body | Trapped air | Isothermal gas | Simulator

OFFICIAL PROBLEM STATEMENT

A hollow cube of side L and weight $W < \rho g L^3$ is initially full of air at the ambient pressure p_a and temperature T_a. At a certain instant, the cube is placed on the surface of a liquid of density ρ. The liquid enters the cube through an orifice in its base until an equilibrium position is reached, characterized by the lengths H_1 and H_2 of the attached figure. Assuming that the evolution of the air inside the cube is isothermal, determine the values of H_1/L and H_2/L, as well as the final pressure inside the cube, p_c.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_3.pdf, p. 1): hollow cube of side L with an orifice in its base; the inner interface is H_1 below the outer free surface and H_2 above the base; air at p_c inside, p_a outside

Source: fluid_statics_3.pdf (page 1, sheet of 27-01-03). Notes.pdf, Chapter 5: Eqs. 5.10, 5.15 and 5.18.

Worked solution

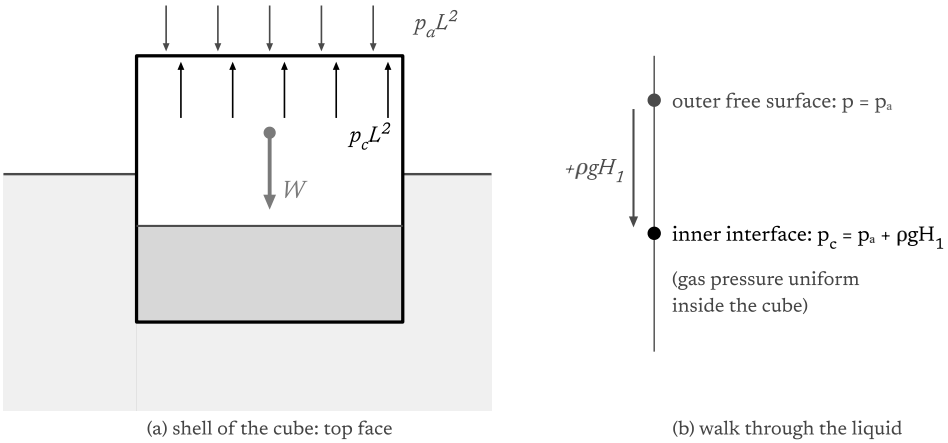
Hypotheses, Data and Unknowns

- Liquid and air at rest; the liquid is incompressible; the outside air is at the uniform pressure p_a.
- The trapped air has a uniform pressure p_c (its weight and the change of its pressure over a height L are negligible, Section 7 of Topic 5) and keeps the temperature T_a: isothermal process, $p\,V = \text{const}$ from the state equation (5.18) at constant T and constant mass.
- Thin walls: the volume of the walls is neglected; W is the weight of the cube alone (the air weight is negligible).
- The cube floats upright with its top face above the outer free surface (checked at the end).
- Data: L, W, ρ, g, p_a. Unknowns: H_1 (depth of the inner interface below the outer free surface), H_2 (height of the inner liquid layer above the base) and p_c.

Symbol	Meaning	SI unit
L	side of the cube	m
W	weight of the cube	N
H_1, H_2	lengths of the figure	m
p_a, p_c	outside pressure, final air pressure inside	Pa

Coordinates and the Free-Body Diagram

Take the vertical axis pointing up. The system for the force balance is the **shell of the cube** alone (not the air, not the liquid inside). Its side walls are vertical, so the pressure on them is horizontal and adds up to zero by symmetry; its base has an orifice and is wetted by the same liquid on both faces at the same depth, so its net pressure force is zero (thin wall). Only the top face carries a net vertical pressure force: $p_a L^2$ downward from the outside air and $p_c L^2$ upward from the trapped air.



(a) Vertical forces on the shell of the cube (thin walls): p_aL^2 down on the top face from outside, p_cL^2 up on it from the trapped air, weight W down; the side walls carry only horizontal forces, which cancel. (b) Walking from the outer free surface down the depth H_1 to the inner interface gives $p_c = p_a + \rho g H_1$

Equilibrium of the Shell

Vertical equilibrium of the shell (upward positive):

$$p_cL^2 - p_aL^2 - W = 0 \quad \Longrightarrow \quad p_c = p_a + \frac{W}{L^2}$$

The Hydrostatic Law between the Two Free Surfaces

The liquid inside the cube is connected with the outside liquid through the orifice, so (5.10) holds through both. Walking from a point of the outer free surface, where $p = p_a$, down a depth H_1 to the level of the inner interface (any path inside the connected liquid gives the same result), the pressure grows by $\rho g H_1$; at the interface it equals the air pressure (Section 1 of Topic 5, no surface tension):

$$p_c = p_a + \rho g H_1$$

Equating the two expressions of p_c:

$$\rho g H_1 = \frac{W}{L^2} \quad \Longrightarrow \quad \frac{H_1}{L} = \frac{W}{\rho g L^3}$$

The hypothesis $W < \rho g L^3$ gives $H_1 < L$.

The Trapped Air: Isothermal Compression

Initially the air fills the whole cube, $V_0 = L^3$, at p_a. At equilibrium it fills the part of the cube above the inner interface, a prism of base L^2 and height $L - H_2$, at p_c. With the same mass and the same temperature, $p\,V$ is constant:

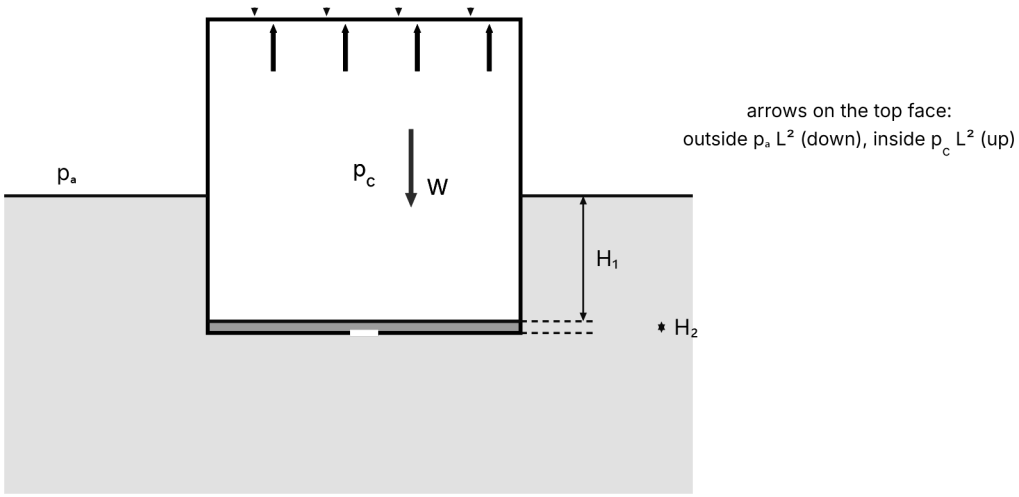
$$p_a L^3 = p_c L^2 (L - H_2) \quad \Longrightarrow \quad \frac{L - H_2}{L} = \frac{p_a}{p_c} \quad \Longrightarrow \quad \frac{H_2}{L} = 1 - \frac{p_a}{p_c} = \frac{p_c - p_a}{p_c}$$

and with $p_c - p_a = W/L^2$ and $p_c = p_a + W/L^2$, multiplying numerator and denominator by L^2:

$$\frac{H_2}{L} = \frac{W/L^2}{p_a + W/L^2} = \frac{W}{p_a L^2 + W}$$

Simulator: the Floating Cube

Simulator: floating hollow cube with trapped air



$H_1/L = W/(\rho g L^3)$	0.4000
$H_2/L = W/(p_a L^2 + W)$	0.0374
$p_c/p_a = 1 + W/(p_a L^2)$	1.0388
Check $(p_c - p_a)L^2 = W$	both equal 0.0388 $p_a L^2$
Top face above the outer surface?	yes: $H_1 + H_2 = 0.437\ L < L$

Answers and Checks

- $H_1/L = W/(\rho g L^3)$, $H_2/L = W/(p_a L^2 + W)$, $p_c = p_a + W/L^2$.
- **Archimedes check.** For the system cube plus trapped air, the displaced liquid is the air below the outer free surface, a prism of base L^2 and height H_1 (from the outer level down to the inner interface). Its weight $\rho g L^2 H_1 = W$ balances the cube, which is the same equation as above.
- **Limiting cases.** $W \to 0$: $H_1 \to 0$, $H_2 \to 0$, $p_c \to p_a$ (a weightless box lets no liquid in). $p_a \to \infty$ (incompressible air): $H_2 \to 0$, the air keeps its volume.
- **Consistency.** The top face is above the outer free surface when $H_1 + H_2 < L$, that is $W/(\rho g L^3) + W/(p_a L^2 + W) < 1$. For water and $L = 1$ m, $p_a/(\rho g L) \approx 10.3$, so $H_2 \ll H_1$: the air is barely compressed and the cube floats almost as a closed box.
- **Dimensions.** $W/(\rho g L^3)$: N / (kg m^{-3} m s^{-2} m^3) = N/N, dimensionless; $W/(p_a L^2)$: N/(Pa m^2) = N/N; W/L^2 in Pa.

- **Check against the official key.** fluid_statics_3.pdf gives $H_1 = W/(\rho g L^2)$, $p_c = p_a + W/L^2$ and $H_2/L = 1 - p_a/p_c = W/(p_a L^2 + W)$, the same results.